LESSON 5.3

Determine the Area of Trapezoids

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can find the area of a trapezoid using the formula A = ½(b1 + b2) × h.

WHY THIS MATTERS Your architecture firm is designing a trapezoidal window for a modern building. The glass supplier charges by the square foot, so the team must calculate the exact area. The window has a top edge of 4 feet, a bottom edge of 8 feet, and a height of 5 feet.

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Trapezoid *Spanish: Trapecio*	A four-sided shape with just one pair of parallel sides.	top base and bottom base
Parallel *Spanish: Paralelas*	Two lines or sides that stay the same distance apart and never meet.	the two bases
Base 1 (b1) *Spanish: Base 1 (b1)*	One of the two parallel sides of a trapezoid.	If the bottom of the trapezoid is 10 ft, then b_1 = 10 ft in the formula A = ½(b_1 + b_2) × h
Base 2 (b2) *Spanish: Base 2 (b2)*	The other parallel side of a trapezoid.	If the top of the trapezoid is 6 ft, then b_2 = 6 ft; both bases are parallel to each other
Height *Spanish: Altura*	The straight-up distance between the two parallel sides.	h, meets bases squarely
Perpendicular *Spanish: Perpendicular*	Two lines that meet to make a square corner (90 degrees).	The dashed height meets the base at a square corner — the small 90° box is what makes it perpendicular, not slanted.

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

Bases 6 cm and 14 cm, height 5 cm, two matching triangles.

1. Rectangle: 6 × 5 = 30 cm².
2. Each triangle base: (14 − 6) ÷ 2 = 4 cm.
3. Triangles: 2 × (½ × 4 × 5) = 20 cm².

Answer: 30 + 20 = 50 cm²

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

Now a smaller trapezoid: bases 3 cm and 5 cm, height 2 cm.

1. _____
2. _____
3. _____
4. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- The two parallel bases are _____ and _____ .
- I decomposed the trapezoid into _____ and _____ .
- The pieces have areas _____ , _____ , and _____ .
- The total area is _____ square units because _____ .

Word bank trapezoid | parallel | base | height | decompose | rectangle | triangle | leftover | add | total | area | square units

WATCH OUT!

A common mistake

A common mistake in Area of Trapezoids is forgetting the ½ in A = ½ × (b1 + b2) × h. After adding the bases and multiplying by the height, students stop there instead of taking half

LESSON 5.3

Determine the Area of Trapezoids

Version A · Supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Complete the table. Show how you found each value.**

Use the formula $A = \frac{1}{2} \times (b_1 + b_2) \times h$ to find the area of each trapezoid.

Trapezoid	Base 1	Base 2	Height	Area
Shape A	4 ft	6 ft	3 ft	
Shape B	8 ft	12 ft	5 ft	

HINT Each row lists Base 1, Base 2, and Height — the Area column is the only blank to fill.

SHOW YOUR WORK

PART 2 Apply it

real-world problems

2 **Circle the letter of the best answer. Show how you know.**

What is the area of a trapezoid with bases 6 cm and 10 cm, and height 4 cm?

(A) 24 sq cm
(B) 32 sq cm
(C) 40 sq cm
(D) 64 sq cm

HINT A trapezoid has two bases (6 cm and 10 cm) plus a height (4 cm) — all three belong in the formula.

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

1. **Name it** *Which measure is asked?*
2. **Write the formula** *Before any numbers.*
3. **Substitute** *Match each letter.*
4. **Unit** *units, sq units, or cubic?*

3 **Circle the letter of the best answer. Show how you know.**

A trapezoid has an area of 45 sq ft, bases of 7 ft and 11 ft. What is the height?

(A) 3 ft
(B) 5 ft
(C) 9 ft
(D) 10 ft

HINT You already know the area (45 sq ft) and both bases (7 ft and 11 ft) — the height is what's missing.

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

1. **Name it** *Which measure is asked?*
2. **Write the formula** *Before any numbers.*
3. **Substitute** *Match each letter.*
4. **Unit** *units, sq units, or cubic?*

4 **Circle the letter of the best answer. Show how you know.**

What is the first step when finding the area of a trapezoid?

(A) Multiply the two bases

(B) Subtract the shorter base from the longer base

(C) Add the two bases together

(D) Divide the height by 2

HINT Look at the trapezoid formula $A = \frac{1}{2} \times (b_1 + b_2) \times h$. Parentheses show what happens first.

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

A trapezoid has bases of 5 in and 9 in, and a height of 6 in. What is its area?

(A) 27 sq in

(B) 30 sq in

(C) 42 sq in

(D) 84 sq in

HINT Circle the two bases (5 in and 9 in) and the height (6 in) — the trapezoid formula uses all three.

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

1. **Name it** *Which measure is asked?*
2. **Write the formula** *Before any numbers.*
3. **Substitute** *Match each letter.*
4. **Unit** *units, sq units, or cubic?*

PART 3 Explain and correct

reasoning

6 Find the mistake. Explain it, then show the correct work.

Find the Missing Half

1. **Read the blueprint** A trapezoid-shaped patio stone has b1 = 4 ft, b2 = 8 ft, and h = 6 ft.
2. **Write the formula** A = ½ × (b1 + b2) × h
3. **Add the bases, then multiply by height** (4 + 8) × 6 = 12 × 6 = 72
4. **State the area** A = 72 sq ft

Which step has the mistake? Explain what went wrong, then show the correct work.

HINT Compare the last step to the formula A = ½ × (b1 + b2) × h — what operation is missing?

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

If you copied this trapezoid, flipped it, and placed it next to the original, what shape would you get? How does this help you find the area?

SENTENCE STARTER The area of the trapezoid is 1/2 x (___ + ___) x ___ = 1/2 x ___ x ___ = ___ square feet.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 5.3

Determine the Area of Trapezoids

Version B · Core practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can find the area of a trapezoid using the formula A = ½(b1 + b2) × h.

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PART 1 Understand the idea

models and meaning

1 **Complete the table. Show how you found each value.**

Complete the table.

Trapezoid	Base 1	Base 2	Height	Area
Window A	8 ft	4 ft	5 ft	
Window B	12 ft	6 ft	4 ft	
Window C	10 ft		8 ft	72 sq ft
Window D		5 ft	6 ft	48 sq ft

SHOW YOUR WORK

2 **Write the letter of the matching item on each line.**

Match each shape to its area.

_____ Trapezoid b_1=10, b_2=6, h=4

_____ Trapezoid b_1=8, b_2=4, h=5

_____ Trapezoid b_1=12, b_2=8, h=4

_____ Triangle b=14, h=6

_____ Parallelogram b=9, h=5

_____ Rectangle 7×3

(A) 32 sq units

(B) 30 sq units

(C) 40 sq units

(D) 42 sq units

(E) 45 sq units

(F) 21 sq units

3 **Write the name of the correct group on each line.**

Calculate the area of each trapezoid, then sort by whether the area is greater than 40 sq units or not.

Groups: **Area > 40 sq units** **Area ≤ 40 sq units**

_____ $b_1=10$, $b_2=6$, $h=4$

_____ $b_1=12$, $b_2=8$, $h=5$

_____ $b_1=6$, $b_2=4$, $h=7$

_____ $b_1=9$, $b_2=7$, $h=6$

_____ $b_1=5$, $b_2=3$, $h=8$

_____ $b_1=14$, $b_2=6$, $h=5$

4 **Mark and label each value on the number line.**

Find the missing height: $b_1=6$, $b_2=10$, $A=48$



SHOW YOUR THINKING

PART 2 Explain and correct

reasoning

5 Find the mistake. Explain it, then show the correct work.

Using the Wrong Height

1. **Read the blueprint** A trapezoid window has b1 = 7 ft and b2 = 13 ft. The slanted side measures 10 ft, and the straight-up height between the bases is 6 ft.
2. **Write the formula** A = ½ × (b1 + b2) × h
3. **Substitute values** A = ½ × (7 + 13) × 10
4. **Calculate** A = ½ × 20 × 10 = 100 sq ft

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

If you copied this trapezoid, flipped it, and placed it next to the original, what shape would you get? How does this help you find the area?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 5.3

Determine the Area of Trapezoids

Challenge · Extension

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can find the area of a trapezoid using the formula A = ½(b1 + b2) × h.

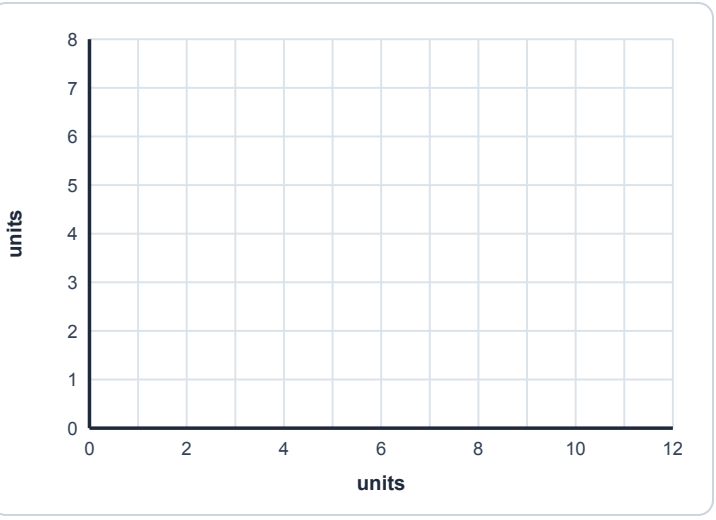
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PART 1 Understand the idea

models and meaning

1 **Plot and label each point on the grid.**

Plot the trapezoid and calculate its area.



SHOW YOUR THINKING

PART 2 Explain and correct

reasoning

2 Find the mistake. Explain it, then show the correct work.

Find the Area Error

1. **Identify the bases and height** $b_1 = 10$ m, $b_2 = 6$ m, $h = 7$ m
2. **Write the formula** $A = ½ \times (b_1 + b_2) \times h$
3. **Substitute values** $A = ½ \times (10 + 6) \times 7$
4. **Calculate** $A = (10 + 6) \times 7 = 112$ sq m

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

3 Answer in complete sentences.

A trapezoid and a parallelogram both have a height of 8 ft. The trapezoid has bases of 5 ft and 11 ft. The parallelogram has a base of 7 ft. Which shape has the greater area? Explain your reasoning using the formulas.

WRITE YOUR ANSWER

EXPLAIN YOUR THINKING

If you copied this trapezoid, flipped it, and placed it next to the original, what shape would you get? How does this help you find the area?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

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